HDOC DAY-54

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BATCH-15

Question

Write a C program to find the maximum sum of a contiguous subarray of size k using a variable-length array (VLA), pointers, and a function named max_subarray_sum().

Display -1 if k is invalid.

Aim

To write a C program to find the maximum sum of a contiguous subarray of size k using pointers and a function.

Algorithm

1. Start.
2. Read the size n of the array.
3. Declare a variable-length array arr[n].
4. Read the array elements and value of k.
5. Pass the array to max_subarray_sum() using a pointer.
6. Check whether k is valid. If not, print -1.
7. Calculate the sum of the first k elements.
8. Slide the window by subtracting the first element and adding the next element.
9. Update the maximum sum whenever a larger sum is found.
10. Display the maximum sum.
11. Stop.

>C PROGRAM:

```c
#include <stdio.h>

void max_subarray_sum(int *arr, int n, int k) {
    if (k <= 0 || k >= n) {
        printf("-1");
        return;
    }

    int sum = 0, max;

    for (int i = 0; i < k; i++)
        sum += arr[i];

    max = sum;

    for (int i = k; i < n; i++) {
        sum += arr[i] - arr[i-k];

        if (sum > max)
            max = sum;
    }

    printf("Maximum subarray sum: %d", max);
}

int main() {
    int n, k;

    printf("Enter size: ");
    scanf("%d", &n);

    if (n <= 0)
        return 0;

    int arr[n];

    printf("Enter elements: ");
    for (int i = 0; i < n; i++)
        scanf("%d", &arr[i]);

    printf("Enter k: ");
    scanf("%d", &k);

    max_subarray_sum(arr, n, k);

    return 0;
}
```

>OUTPUT:

```
zsh: corrupt history file /home/priyadarshna/.zsh_history
┌──(priyadarshna㉿kali)-[~]
└─$ gcc day54.c -o day54

┌──(priyadarshna㉿kali)-[~]
└─$ ./day54
Enter size: 6
Enter elements: 2 1 5 1 3 2
Enter k: 3
Maximum subarray sum: 9

┌──(priyadarshna㉿kali)-[~]
└─$ ./day54
Enter size: 5
Enter elements: 2 3 4 1 5
Enter k: 2
Maximum subarray sum: 7

┌──(priyadarshna㉿kali)-[~]
└─$ ./day54
Enter size: 3
Enter elements: 1 2 3
Enter k: 4
-1

┌──(priyadarshna㉿kali)-[~]
└─$
```